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Anti-Müllerian hormone: a novel biomarker for aggressive prostate cancer? Emerging evidence from a prospective study of radical prostatectomies

Stavros Kontogiannis¹, Georgios Markantes², Maria Stamou³, Michail Tsagkarakis¹, Iirini Mamali², Konstantinos Giannitsas¹, Petros Perimenis¹, Neoklis Georgopoulos², Anastasios Athanasopoulos¹

¹Urology Department, Patras University Hospital, Patras, Greece

²Endocrinology Department, Patras University Hospital, Patras, Greece

³Harvard Reproductive Sciences Center, Massachusetts General Hospital, Boston, MA, USA

Abstract

Purpose—Prostate cancer patients are a heterogeneous group as regards the aggressiveness of the disease. The relationship of steroid hormones with the aggressiveness of prostate cancer is unclear. It is known that the anti-Müllerian hormone (AMH) inhibits prostate cancer cell lines in vitro. The aim of this study is to investigate the relationship of AMH and steroid hormones with the aggressiveness of prostate cancer.

Methods—This was a prospective study of consecutive radical prostatectomy patients. We measured the following hormones: total testosterone, sex hormone-binding globulin, albumin, luteinizing hormone, follicle-stimulating hormone, estradiol, dehydroepiandrosterone sulfate, androstenedione, and AMH. The minimum follow-up after radical prostatectomy was 5 years. For the aggressiveness of prostate cancer, we considered the following three variables: post-operative Gleason score (GS) ≥ 8 , TNM pT3 disease, and prostate-specific antigen (PSA) biochemical recurrence (BCR).

Results—In total, 91 patients were enrolled. The mean age and PSA were 64.8 years and 9.3 ng/dl, respectively. The median post-operative GS was 7. Low AMH blood levels were correlated with higher post-operative GS ($p = 0.001$), as well as with PSA BCR ($p = 0.043$). With pT3 disease, only albumin was (negatively) correlated ($p = 0.008$). ROC analysis showed that AMH is a good predictor of BCR (AUC 0.646, 95% CI 0.510–0.782, $p = 0.043$); a cutoff value of 3.06 ng/dl had a positive prognostic value of 71.4% and a negative prognostic value of 63.3% for BCR. Cox regression analysis showed that AMH is a statistically significant and independent prognostic marker for BCR ($p = 0.013$). More precisely, for every 1 ng/ml of AMH rise, the probability for

[✉] Stavros Kontogiannis, stavroskontogiannis@gmail.com.

Declarations

Ethical approval Our study was performed according to the Declaration of Helsinki and an approval from the Ethics Committee of our University Hospital was obtained.

Informed consent All patients signed an informed consent before participating in the study.

Conflict of interest The authors declare no competing interests.

PSA BCR decreases by 20.8% (HR = 0.792). Moreover, in Kaplan–Meier analysis, disease-free survival is more probable in patients with AMH $\geq$ 3.06 ng/ml ($p = 0.004$).

Conclusions—Low AMH blood levels were correlated with aggressive prostate cancer in this radical prostatectomy cohort of patients. Therefore, AMH could be a prognostic biomarker for the aggressiveness of the disease.

Keywords

Anti-Müllerian hormone (AMH); Prostate cancer; Biomarker; Radical prostatectomy

Introduction

Prostate cancer is a heterogeneous group of malignancies with variable aggressiveness and prognosis [1, 2]. After radical prostatectomy, some patients will be disease-free for many years, while others will have aggressive disease and poor prognosis [3, 4]. The latter group of patients may have residual disease right after surgery or they may develop prostate-specific antigen (PSA) biochemical recurrence (BCR) some years later [5, 6]. To achieve improved and prompt diagnosis and treatment of prostate cancer, it is important to identify which patients with prostate cancer will develop aggressive disease even at the early stages of the disease.

The aggressiveness of prostate cancer is usually predicted through estimation of the risk for PSA BCR by the following two methods: (i) using the Gleason score (GS) and (ii) measuring the PSA before surgery. High PSA and GS increase the risk for PSA BCR and hence, for aggressiveness of prostate cancer [7]. The GS is reported by a pathologist in each case, and although it is based on published guidelines [8], it is still a subjective estimation. For example, prior studies have shown that two different pathologists can score the same specimen in a different way [9, 10]. Being able to use a more objective variable for aggressiveness at the time of diagnosis would be extremely valuable for both patients and the treatment team of healthcare professionals.

Hormones are objective variables measured by the laboratories and do not depend on subjective estimations like the GS. They are continuous variables and the measurements are accurate. In addition, a particular hormonal value reflects the entire prostate, whereas a pathology report is typically based on a portion of the gland. Testicular androgens (mainly testosterone) have been found to be correlated with aggressive prostate cancer [11–14]. Adrenal androgens are also investigated [15–17]. Nevertheless, no definite conclusion has been reached on the role of testicular or adrenal sex steroids in prostate cancer aggressiveness.

Anti-Müllerian hormone (AMH) is even less investigated concerning its role in prostate cancer aggressiveness. AMH is a glycoprotein hormone which is secreted by Sertoli cells. In the male fetus, AMH inhibits the development of the female reproductive tract, that is, the paramesonephric or Müllerian ducts [18]. In addition, as a member of the transforming growth factor beta (TGF- β) super-family, AMH has been shown to play an anti-tumor role [19]. There is already some evidence that AMH inhibits prostate cancer cells in vitro [20–

22]. The objective of this study is to investigate the role of AMH, testicular, and adrenal sex steroids in the aggressiveness of prostate cancer in patients undergoing prostatectomy.

Patients and methods

Patient population

This was a prospective study. We enrolled consecutive patients who underwent radical prostatectomy for clinically localized prostate cancer in our department (from January 2014 until June 2015). All patients had a negative bone scan and a negative computed tomography before surgery. Patients under hormonal treatment were excluded from the study.

The follow-up period was until August 2020, the end of the study period. The median follow-up was 5.5 years and the minimum follow-up was 5 years. We set the minimum follow-up at 5 years as most patients show a PSA BCR during the first 5 years after surgery [23, 24].

Study variables

Hormonal measurements—We measured all the hormones once on the morning before surgery in fasted patients. We measured the following steroid hormones: total testosterone, luteinizing hormone (LH), follicle-stimulating hormone (FSH), and estradiol. To calculate the free and bioavailable testosterone, we also measured albumin and sex hormone-binding globulin (SHBG). Free and bioavailable testosterone were calculated via the online calculator of the University Hospital of Ghent [25]. To investigate the role of the adrenal androgens in prostate cancer, we also measured dehydroepiandrosterone-sulfate (DHEA-S) and androstenedione. Lastly, we measured AMH from the same blood samples (electrochemiluminescence immunoassay, ROCHE, HITACHI) only once again with the goal to investigate its role in prostate cancer prognosis.

Histopathological features before and after surgery—We collected the histopathological features of the patients before and after surgery. Before surgery, we noted the PSA before the prostate biopsy and the prostate biopsy GS. Through clinical examination prior to surgery, we calculated the clinical tumor stage according to the TNM classification system [26]. According also to the European Association of Urology Guidelines, we additionally categorized our patients as having low, medium, and high risk of prostate cancer recurrence [7]. In medium and high-risk patients, limited lymphadenectomy (external iliac and obturator lymph nodes) was performed. After surgery, we noted the pathology GS and the pathological stage of the disease according to the TNM classification system and the presence or absence of positive lymph nodes.

Follow-up procedure—The first PSA measurement was obtained within the first 3 months after surgery for all patients. If a patient had a PSA ≥ 0.2 ng/ml at 3 months after surgery, he was excluded from further follow-up. For the rest of the patients who had an initial PSA < 0.2 ng/ml post-operatively, we obtained a PSA measurement every 3 months during the first 2 years and every 6 months thereafter. According to the European Association of Urology definition of PSA BCR [27], if the PSA was found to be > 0.2 ng/ml

with two consecutive increases, we considered this patient as having BCR. Patients who had a PSA < 0.2 ng/ml during the whole follow-up period composed the disease-free survival group.

Prostate cancer aggressiveness variables—For our analysis, we considered three variables as signs of cancer aggressiveness, namely, two post-operative histopathological variables and one follow-up variable. The histopathological variables of aggressiveness consisted of high pathology GS (GS > 8) and pT3 disease (extracapsular disease). The follow-up variable of aggressiveness was PSA BCR after surgery.

Statistical analysis

Data were analyzed using IBM SPSS Statistics for Windows, version 26.0 (IBM Corp., Armonk, NY, USA). Parameters were tested for normality with the Kolmogorov–Smirnov test. Continuous variables are presented using mean (SD) or median (IQR), while categorical variables are presented using absolute and relative (%) frequencies. Comparisons between subgroups were performed using the independent samples *t*-test or one-way ANOVA (for normally distributed data) and the Mann–Whitney *U* test or the Kruskal–Wallis *H* test (for non-normally distributed data). Correlations were estimated using Pearson or Spearman correlation tests, as appropriate. Receiver operating characteristics (ROC) curve and area under the curve (AUC) were calculated to assess the ability of AMH to predict PSA BCR. The maximum sum of sensitivity and specificity was used to determine the optimal threshold value for AMH. Cox’s proportional hazards model was performed to evaluate the effect upon time of potential predictors of disease BCR (expressed as HR). The disease-free survival of the patient subgroups was estimated by means of the Kaplan–Meier method, and differences between curves were analyzed by the log-rank test. All tests were 2-tailed and a *p*-value of less than 0.05 was considered significant.

Results

Ninety-one patients were enrolled in the study. The mean (SD) age was 64.8 (6.7) years and the median (IQR) PSA was 7.5 (3.9) ng/dl. Almost half of our patients had a biopsy GS of 7 (49.5% of the cohort). Concerning the clinical staging, most of them had T1c clinical disease (34.1%). Only 24.2% of the patients belonged to the EAU low-risk group. Most of them belonged to the EAU medium-risk group, while a significant percentage (28.6%) belonged to the high-risk EAU group. The pathological reports revealed a high percentage of GS 7 (56%), as expected, as well as a high percentage of GS 8–10 (23.1%). Limited lymphadenectomy was performed in 69 patients with only seven of them having positive lymph nodes. Concerning the follow-up period, three patients were lost to follow-up and 23 patients had PSA > 0.2 ng/ml right after surgery and were excluded from further follow-up. Of the remaining 65 patients, 31 patients showed PSA BCR and 34 patients had disease-free survival. All the patient characteristics are presented in Table 1. In addition, the means and standard deviations of all the serum measurements are also displayed in Table 1.

Concerning the correlations of the serum measurements and the variables of aggressive prostate cancer, we found that AMH was negatively correlated with the post-operative GS (Spearman’s $\rho = -0.337$, $p = 0.001$), i.e., low AMH blood levels were correlated

with high post-operative GS. When patients were stratified into three groups according to their post-operative GS (low = 6, intermediate = 7, high = 8–10), there was a statistically significant difference in AMH levels between the groups; post hoc analysis revealed that this difference was due to significantly lower AMH levels in the high GS compared to the low GS group (2.80 ng/ml vs 5.69, $p = 0.005$). Furthermore, median AMH levels were lower in the patients who had PSA BCR compared with those who did not (3.31 ng/ml vs 4.91 ng/ml, respectively, $p = 0.043$). Finally, AMH levels were not correlated with the presence of pT3 (extracapsular) disease (Table 2). All other measured hormonal parameters were not significantly correlated with the aggressiveness variables studied. However, albumin showed a negative correlation with the post-operative GS (Spearman's $\rho = -0.258$, $p = 0.014$), while median albumin levels were lower in patients with compared to those without T3 disease (4.1 g/dl vs 4.3 g/dl, $p = 0.008$). Albumin was not correlated with PSA BCR (Table 2).

Considering the novel associations of serum AMH values with the cancer's aggressiveness, we conducted a series of other statistical tests to investigate the role of AMH in predicting PSA BCR.

First, we performed a ROC analysis which demonstrated that AMH had a close to fair predictive value for PSA BCR (AUC 0.646, 95% CI 0.510–0.782, $p = 0.043$, Fig. 1). An AMH level of 3.06 ng/dl was identified as the optimal threshold to discriminate between PSA BCR and disease-free survival. The positive predictive value (PPV) of this cutoff was 71.4% and the negative predictive value (NPV) was 63.6%.

Subsequently, we conducted a Cox regression analysis to evaluate the prognostic value of both AMH and the presence of pT3 disease in predicting PSA BCR. The analysis showed that both AMH and pT3 disease are important and independent predictors of PSA BCR. As shown in Table 3, for every 1 ng/ml increase in AMH, the probability of PSA BCR is decreased by 20.8% (HR = 0.792). It is very important that this occurs regardless of the disease stage (either in pT2 or pT3 patients). Furthermore, the presence of pT3 increases the probability of BCR 3.447 times compared to the absence of pT3, regardless of AMH levels.

Finally, we divided all 65 patients who had PSA < 0.2 ng/ml after surgery into two groups, namely, those with AMH < 3.06 ng/ml and those with AMH $\geq$ 3.06 ng/ml. Disease-free survival (DFS) of the two groups was estimated using the Kaplan–Meier method. We found that DFS was significantly higher in patients with AMH $\geq$ 3.06 ng/ml (log-rank test $p = 0.004$, Fig. 2a). Similarly, when pT2 or pT3 patients were studied separately, AMH $\geq$ 3.06 ng/ml was correlated with increased DFS too (log-rank test $p = 0.047$ & 0.024, respectively, Fig. 2b and c).

Discussion

This is the first study, to our knowledge, to show an association of AMH blood levels with prostate cancer aggressiveness after radical prostatectomy. Low serum AMH levels are correlated with features of aggressiveness both right after surgery but also years after surgery, as shown in our cohort with a minimum follow-up of 5 years. More specifically, low

AMH blood levels (< 3.06 ng/dl) were associated both with higher post-operative GS as well as with a higher rate of PSA BCR. Another important finding is that AMH is an independent predictor of PSA BCR. Low AMH levels (< 3.06 ng/dl) can predict PSA BCR irrespective of the pathological stage of the disease (Fig. 2b and c). Based on this analysis, we may preoperatively be able to predict if the patient is going to have BCR or not, irrespective of whether he will have pT2 or pT3 disease post-operatively.

Our results could be translated into clinical practice in two ways. First, AMH is accurately and objectively measured in the serum as opposed to a subjective evaluation of the GS of the specimen by a pathologist. This makes AMH measurement an important and useful biomarker in the everyday practice of urologists. It is crucial to know whether a patient has low serum levels of AMH before radical prostatectomy for the following two reasons: (i) clinicians need to inform patients with low AMH levels about the probable aggressiveness of the disease and (ii) they must follow them up closely for PSA BCR. Second, since human recombinant AMH has been made available [28, 29] and has been shown to induce apoptosis in human prostatic cancer cells [20–22], its potential role as a novel treatment for prostate cancer should be studied.

Prior studies have investigated the possible role of AMH as a screening tool in prostate cancer. The study of Sklavos et al. [30] focused on circulating AMH in relation to risk of prostate cancer. It investigated prediagnostic AMH levels in relation to prostate cancer in 998 cases and 999 controls and no association was observed between AMH quartiles and risk of prostate cancer (OR Q4 vs Q1 = 1.15, 95% CI: 0.89, 1.48). However, Verdiesen et al. [19] noted that in the latter study, there was no examination of the potential effect of age at blood draw. Furthermore, Wang et al. [31] explored the role of the tumor mutational burden in prostate cancer patients and identified a role of AMH in overall survival and cancer recurrence. The study by Wang et al. was a retrospective analysis of data from The Cancer Genome Atlas (TCGA) database. The data concerned 188 prostate cancer specimens (from 132 living and 56 deceased patients). The authors found that AMH gene expression was increased in prostate cancer specimens of high T and N stage. Furthermore, patients with low AMH expression had longer survival times compared to patients with higher expression. Moreover, the DFS analysis indicated that elevated AMH expression favored disease progression in prostate cancer. It must be noted that the authors explored AMH gene expression in the prostate cancer specimen and did not measure AMH in the blood: AMH gene expression in each tissue is not necessarily correlated with AMH blood levels.

In contrast to prior efforts, our study is, to the best of our knowledge, the first prospective study to focus on the correlation of AMH with the aggressiveness of an already diagnosed prostate cancer. Our study indicated that AMH may have a protective role in prostate cancer. But how is this mediated? AMH can play a role as an anti-tumor agent [32, 33]. As a member of the TGF- β super-family, it induces apoptosis and inhibits growth in cells [34, 35]. More specifically, there is evidence that AMH exerts apoptotic action in prostatic cancer cell lines in vitro [20–22]. AMH acts by binding to its receptor and AMH receptor II has been found to be over-expressed in prostate cancer cells [36, 37]. Thus, AMH might also induce apoptosis in prostate cancer in vivo. There is also some evidence of the role of AMH in the immune system, the activation of which has a central role in the defense against

cancer growth. Tran et al. [38] showed that AMH works through a protein complex that controls cell survival, nuclear factor-kappa B (NF- κ B). If the NF- κ B-dependent mechanism fails to be activated by AMH, then the cancer cells can persist despite the presence of immune cells [39].

Albumin was linked with pT3 disease and high GS in our study, which is also an important finding. This result is supported by other cancer studies, where low albumin was correlated with malnutrition and worse survival [40–43]. In addition, in our study, no other steroid hormone was correlated with prostate cancer aggressiveness. This comes in contradiction to some other studies in the field [11–13]. However, because of the heterogeneity of the disease, testosterone alone could not easily be correlated with prostate cancer aggressiveness [43]. Therefore, the role of pretreatment circulating hormones in determining prostate cancer aggressiveness is still unclear.

The main limitations of this study are the relatively small sample size and the one-center design. More studies, particularly of multicenter design, with larger samples must be conducted in the future in order to clearly identify the correlation of AMH with prostate cancer aggressiveness in patients undergoing prostatectomy. Another limitation is that we measured AMH only once before surgery in order to determine its predictive value of aggressive prostate cancer before surgery. It would have been better if we also had AMH measurements after surgery. Although AMH levels are not expected to vary significantly in a given individual within a short time window, fluctuations might exist; therefore, multiple measurements could provide more robust data to calculate cutoff values of AMH as an aggressive prostate cancer predictor. Despite these limitations, the strengths of the study are its prospective nature and minimum follow-up of 5 years. The fact that we used blood samples for measuring AMH could help in the widespread application of AMH as a potential biomarker for prostate cancer aggressiveness.

In conclusion, serum AMH levels < 3.06 ng/ml were correlated with aggressive prostate cancer in this prospective cohort of patients who underwent radical prostatectomy. These low serum AMH levels were correlated with higher post-operative GS and higher probability of PSA BCR. Therefore, AMH might be a new biomarker for prediction of aggressive prostate cancer in patients undergoing prostatectomy.

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Data Availability

All the data are available upon request.

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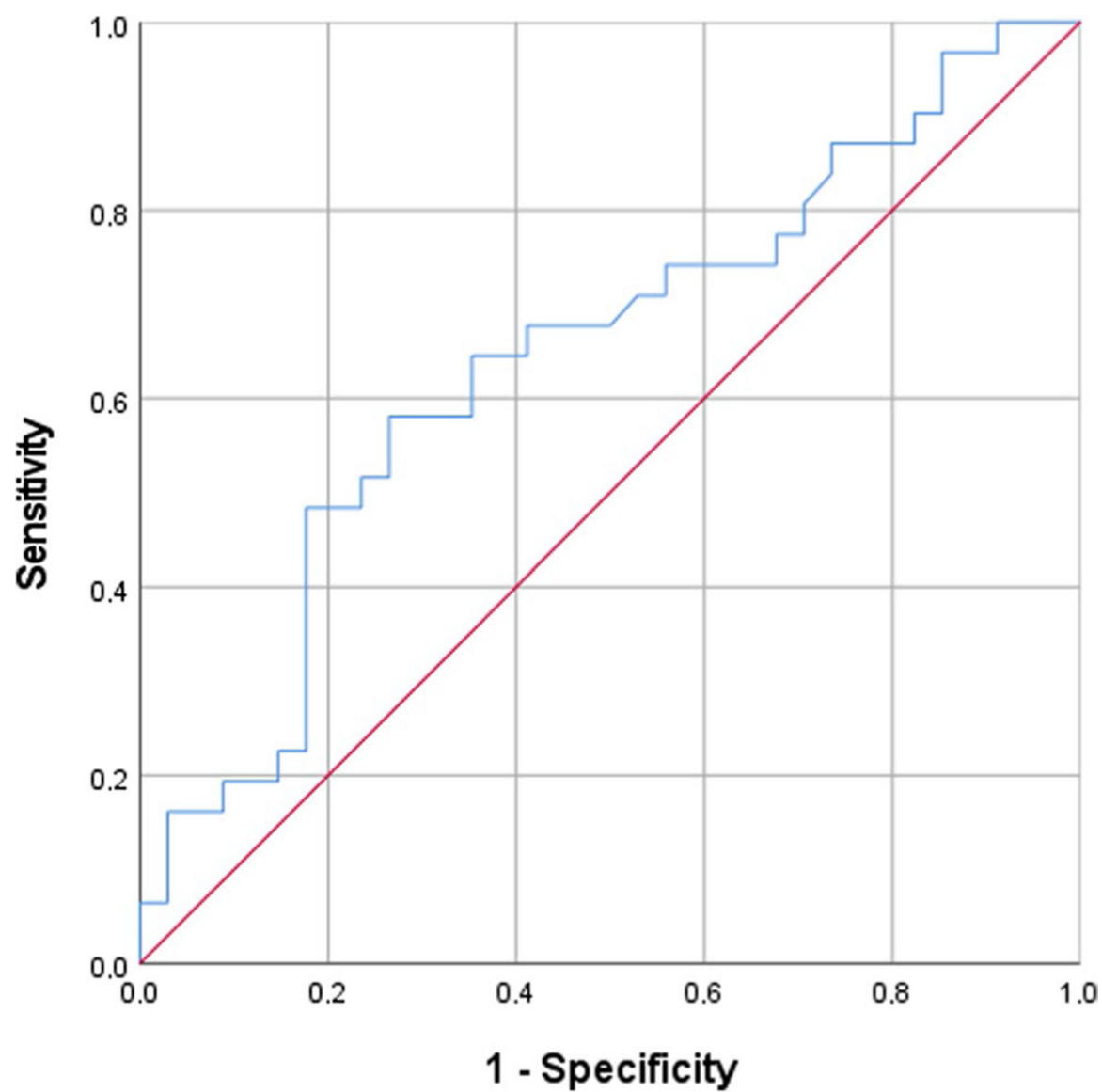
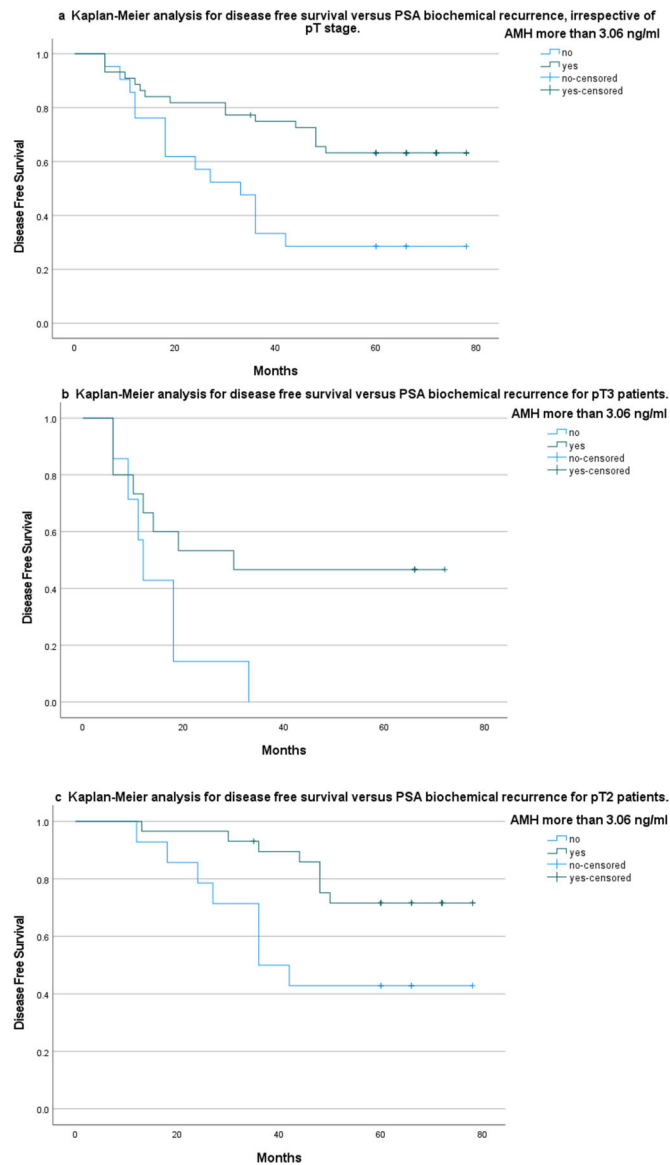


Fig. 1.
ROC curve for PSA biochemical recurrence, AMH cutoff 3.06 ng/ml

**Fig. 2.**

a Kaplan–Meier analysis for disease-free survival versus PSA biochemical recurrence, irrespective of Pt stage, AMH more than 3.06 ng/ml. **b** Kaplan–Meier analysis for disease-free survival versus PSA biochemical recurrence for pT3 patients, AMH more than 3.06 ng/ml. **c** Kaplan–Meier analysis for disease-free survival versus PSA biochemical recurrence for pT2 patients, AMH more than 3.06 ng/ml

Patient characteristics and serum measurements in our cohort. Continuous variables are presented as mean $\pm$ SD (normally distributed) or median (IQR) (non-normally distributed), while categorical variables are presented using absolute and relative (%) frequencies

Table 1

Age (years)	64.8 $\pm$ 6.7	Albumin (g/dl)	4.2 (0.4)
Before surgery		SHBG (nmol/l)	49.9 $\pm$ 19.0
PSA (ng/dl)	7.5 (3.9)	Testosterone (ng/dl)	452.9 $\pm$ 83.3
		Free testosterone (ng/dl)	7.4 (3.6)
Biopsy Gleason score	6 34	Bioavailable testosterone (ng/dl)	166.0 (75.0)
	7 45		
	8–10 12		
Clinical T stage	T1c 31	DHEA-S (μ g/dl)	135.9 (103.9)
	T2a 20		
	T2b 23		
	T2c 17		
EAU risk group	Low 22	Androstenedione (ng/ml)	2.8 (1.6)
	Medium 43		
	High 26		
After surgery		Estradiol (pg/ml)	24.3 (12.0)
Pathological Gleason score	6 21	LH (mIU/ml)	4.1 (3.9)
	7 49		
	8–10 21		
Pathological T stage	pT2 51	FSH (mIU/ml)	6.2 (4.3)
	pT3 40		
Lymph nodes	Nx 22	AMH (ng/ml)	4.2 $\pm$ 2.3
	N0 62		
	N1 7		
Follow-up			
PSA > 0.2 ng/ml right after surgery	23		25.2
PSA biochemical recurrence	31		34.1
Disease-free survival	34		37.4
Lost to follow-up	3		3.3

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Table 2
Correlation of serum measurements and post-operative Gleason score, pT3 disease, and PSA biochemical recurrence

	Post-operative Gleason score		pT3 disease		PSA biochemical recurrence	
	Spearman's rho	p-value	Yes	No	Yes	No
AMH (ng/ml)	- 0.337	0.005 *	3.95 (3.35)	4.42 (3.25)	3.31 (3.39)	4.91 (2.78)
Albumin (g/dl)	- 0.258	0.021 *	4.10 (0.40)	4.30 (0.40)	4.10 (0.70)	4.30 (0.25)
SHBG (nmol/l)	0.033	0.674	48.3 ± 19.5	51.2 ± 18.6	50.8 (21.2)	51.6 (19.2)
Testosterone (ng/dl)	0.065	0.675	440.1 ± 196.8	463.1 ± 173.4	464.7 (188.5)	459.9 (173.9)
Free testosterone (ng/dl)	0.123	0.238	7.40 (17.23)	7.44 (3.83)	6.98 (3.30)	7.40 (3.39)
Bioavailable testosterone (ng/dl)	0.167	0.182	164.0 (81.5)	171.0 (75.0)	161.0 (62.0)	168.5 (80.0)
Estradiol (pg/ml)	0.062	0.715	23.69 (42.12)	24.81 (14.20)	24.23 (15.24)	24.88 (13.18)
DHEA-S (µg/dl)	0.009	0.919	136.9 (99.5)	124.6 (122.1)	136.90 (74.80)	137.35 (118.38)
Androstenedione (ng/ml)	0.132	0.455	2.90 (1.90)	2.70 (1.30)	3.00 (2.10)	2.70 (1.30)
LH (mIU/ml)	0.040	0.928	4.00 (4.35)	4.10 (3.10)	4.80 (4.60)	3.95 (3.03)
FSH (mIU/ml)	0.105	0.316	6.65 (5.38)	5.90 (3.90)	6.80 (4.70)	5.85 (4.92)

* Significant at the 0.05 level
For normally distributed variables, values are presented as mean ± SD and for non-normally distributed variables as median (IQR)

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Table 3

Cox regression analysis for AMH and pT3 concerning PSA biochemical recurrence

	Coefficient	SE	p value	Hazard Ratio	95% CI for hazard ratio
AMH	-0.233	0.094	0.013*	0.792	0.659–0.952
pT3	1.238	0.370	<0.001**	3.447	1.668–7.126